# QCQS SPACES OVER QUASI-SEPARATED BASES ARE LIMITS OF FINITE PRESENTATION ONES

THE STACKS PROJECT AUTHORS

*Editorial note.* Counted span is the author's own proof environment, source lines 2871-2902, following the statement at 2858-2869.

*Editorial note.* No omissions; the three-factor affineness computation is displayed as a composition.

*Allowed standard prerequisites (editorial).*

- Approximation of morphisms. lemma-approximate-morphism (same chapter, card L26305, cited at ~2875-2877). (X -> Y) = lim (X_i -> Y_i) over Z with affine transitions.
- Limits commute with limits. Categories, Lemma categories-lemma-colimits-commute (cross-chapter, cited at ~2878-2879). X = lim X_i x_{Y_i} Y.
- Affineness of the transition morphisms. Morphisms of Spaces, Lemmas spaces-morphisms-lemma-fibre-product-after-map, spaces-morphisms-lemma-base-change-affine, spaces-morphisms-lemma-composition-affine (cross-chapter, cited at ~2887-2893). Transitions are a closed immersion after a base change, hence affine.
- Finite presentation of the base changes. Morphisms of Spaces, Lemmas spaces-morphisms-lemma-noetherian-finite-type-finite-presentation, spaces-morphisms-lemma-finite-presentation-permanence, spaces-morphisms-lemma-base-change-finite-presentation (cross-chapter, cited at ~2894-2901). X_i x_{Y_i} Y -> Y is of finite presentation.

*Source reference.* Original reference 002M: focused-seeds/geometry/S9-spaces-limits-relative-approximation/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

*Source reference.* Original lemma 0GS3: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 2796–2796. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

*Source reference.* Original reference 03WI: focused-seeds/geometry/S9-spaces-limits-relative-approximation/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

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*Date*: stacks-project:spaces-limits.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

*Source reference.* Original reference 03XR: focused-seeds/geometry/S9-spaces-limits-relative-approximation/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

*Source reference.* Original reference 03WH: focused-seeds/geometry/S9-spaces-limits-relative-approximation/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

*Source reference.* Original reference 03KN: focused-seeds/geometry/S9-spaces-limits-relative-approximation/ref-tags.txt, source lines 9–9. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

*Source reference.* Original reference 05WT: focused-seeds/geometry/S9-spaces-limits-relative-approximation/ref-tags.txt, source lines 10–10. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

*Source reference.* Original reference 06G4: focused-seeds/geometry/S9-spaces-limits-relative-approximation/ref-tags.txt, source lines 11–11. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

**Lemma 0.1.** *Let $S$ be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over $S$. Assume that*

(1) *$X$ is quasi-compact and quasi-separated, and*
(2) *$Y$ is quasi-separated.*

*Then $X = \lim X_i$ is a limit of a directed inverse system of algebraic spaces $X_i$ of finite presentation over $Y$ with affine transition morphisms over $Y$.*

*Proof.* Since $|f|(|X|)$ is quasi-compact we may replace $Y$ by a quasi-compact open subspace whose set of points contains $|f|(|X|)$. Hence we may assume $Y$ is quasi-compact as well. By Lemma 0GS3 we can write $(X \to Y) = \lim(X_i \to Y_i)$ for some directed inverse system of morphisms of finite type schemes over $\mathbf{Z}$ with affine transition morphisms. Since limits commute with limits (Categories, Lemma 002M) we have $X = \lim X_i \times_{Y_i} Y$. For $i \geq i'$ the transition morphism $X_i \times_{Y_i} Y \to X_{i'} \times_{Y_{i'}} Y$ is affine as the composition

$$X_i \times_{Y_i} Y \to X_i \times_{Y_{i'}} Y \to X_{i'} \times_{Y_{i'}} Y$$

where the first morphism is a closed immersion (by Morphisms of Spaces, Lemma 03KN) and the second is a base change of an affine morphism (Morphisms of Spaces, Lemma 03WI) and the composition of affine morphisms is affine (Morphisms of Spaces, Lemma 03WH). The morphisms $f_i$ are of finite presentation (Morphisms of Spaces, Lemmas 06G4 and 05WT) and hence the base changes $X_i \times_{f_i, Y_i} Y \to Y$ are of finite presentation (Morphisms of Spaces, Lemma 03XR). □